Edward Y. Chang
Department of Electrical & Computer Engineering
Department of Computer Science (affiliated)
University of California
Santa Barbara, California 93106
Phone: (805) 893-2917
Email: echang@ece.ucsb.edu

EDUCATION

Ph.D.	Stanford University, Electrical Engineering (Advisor: Hector Garcia-Molina)	1995-99
M.S.	Stanford University, Computer Science	1992-94
M.S.	University of California, Berkeley, Industrial Engineering & Operation Research	1984-85

AWARDS

NSF Career Award	2002
IBM Faculty Partnership Award	2000, 2001, 2002

EXPERIENCE

Assistant Professor, University of California, Santa Barbara	1999-present
Research Assistant, Stanford University	1995-99
Staff Engineer, Sun Microsystems, Mountain View, CA	1994-95
Principal Software Engineer, Digital Equipment Corp. Palo Alto, CA	1987-94

SELECTED PATENTS

"Virtual IO: Preemptible Disk Access,"
E. Chang, with University of California., 2002.
"DPF: Dynamic Partial Function,"
E. Chang, with VIMA Technologies Inc., 2002.
"Perception-Based and Personalizable Image Retrieval,"
E. Chang, with University of California, 2001.
"Personalizable Interactive Digital TV,"
E. Chang, with University of California, 2001.
"Methods for Categorizing and Organizing WWW Document Collections,"
K. Vu, W. Li and E. Chang, with NEC Research Labs., 2000.

PROFESSIONAL ACTIVITIES

Associated PC Chair for ACM MM 2002/2003
PC Member for IEEE MM 2002/2003, ICDE 2002, WISE 2002, PAKDD 2002, WAIM 2002, IS&T/SPIE 2002/2003, IEEE CBAIVL 2000/2001.
Reviewer for ACM MM, SIGMOD, PODS, TODS, TOIS; IEEE ICDE, TKDE, TMM, etc.
Local Arrangement Chair for ACM SIGMOD 2001.

SELECTED INVITED TALKS

- "*Perception-based Image Retrieval,*" Stanford Venture Seminar, March 2002.
- "*Dynamic Partial Function,*" UC Berkeley Digital Library Seminar, February 2002.
- "*Multimedia Data Mining,*" HP Labs., December 2001; January 2002.
- "*Measuring and Learning Perceptual Similarity,*" UCLA, October 2001.
- "*Measuring and Learning Perceptual Similarity,*" INRIA-Rocquencourt, September 2001.
- "*Learning Query Concepts via Intelligent Sampling,*" IBM Almaden, September 2001.
- "*Learning Query Concepts via Intelligent Sampling,*" IBM T.J. Watson, August 2001.
- "*Personalizable Interactive Digital VCR,*" SONY Research, April 2001.
- "*On Managing Continuous Media Data,*" Compaq Western Research Lab, September 2000.

FIVE RELATED PUBLICATIONS

- *"MEGA: The Maximizing Expected Generalization Algorithm for Learning Complex Query Concepts,"* E. Chang and B. Li, in ACM Trans. on Information Systems, 2003 (to appear).
- *"The Discovery of A Percpeptual Distance Function for Measuring Image Similarity,"* B. Li, E. Chang, and Y. Wu, ACM Multimedia Journal Special Issue on Content-Based Image Retrieval, 2003 (to appear).
- *"Content-based Soft Annotation for Multimodal Image Retrieval Using Bayes Point Machines,"* E. Chang, K. Goh, G. Sychay, and G. Wu, IEEE Transactions on Circuits and Systems for Video Technology Special Issue on Conceptual and Dynamical Aspects of Multimedia Content Description (to appear), 2003.
- *"DynDex: A Dynamic and Non-metric Space Indexer,"* K. Goh, B. Li, and E. Chang, in Proc. of ACM Conference on Multimedia, pp.466-475, December 2002.
- *"Support Vector Machine Active Learning For Image Retrieval,"* S. Tong and E. Chang, in Proc. of the 9^{th} ACM International Conference on Multimedia, pp.107-118, Ottawa, October 2001.

FIVE OTHER PUBLICATIONS

- *"Design, Analysis, and Implementation of Preemptible IO,"* Z. Dimitrijevic, R. Rangaswami, and E. Chang, in Proc. USENIX Conference on File and Storage Technologies (FAST), April 2003 (to appear).
- *"MEMS-based Disk Buffer for Streaming Media Servers,"* R. Rangaswami, Z. Dimitrijevic, E. Chang, and K. Schauser in Proc. of IEEE International Conference on Data Engineering, March 2003 (to appear).
- *"Virtual IO: Preemptible Disk Access,"* Z. Dimitrijevic, R. Rangaswami, and E. Chang, in Proc. of ACM International Conference on Multimedia, December 2002.
- *"Approximate Similarity Queries in High-Dimensional Spaces,"* C. Li, E. Chang, H. Garcia-Molina, and G. Wiederhold, IEEE Transactions on Knowledge and Data Engineering, Volume 14, Number 4, pp.792-808, July/August 2002.
- *"Answering Queries with Useful Bindings,"* C. Li and E. Chang, ACM Transaction on Databases, Volume 2, Number 3, pp.313-343, September 2001.

INVITED PAPERS

- *"On Learning Perceptual Distance Functions for Image Retrieval,"* B. Li and E. Chang, to appear in Proc. of IEEE International Conference on Acoustics, Speech and Signal Processing, Orlando, May 2002.
- *"High-dimensional Visual Data Indexing,"* K. Goh and E. Chang, to appear in Proc. of the 6^{th} International Conference on Visual Databases, Australia, May 2002.
- *"Learning and Measuring Perceptual Similarity,"* E. Chang, in Proc. of the 1^{st} NSF/INRIA/Berkeley/IBM Workshop on Content-based Multimedia Information Retrieval, Paris, September, 2001.

COLLABORATORS & GRADUATE STUDENTS

Hector Garci-Molina (graduate advisor), Chen Li (collaborator), Simon Tong (collaborator), James Wang (collaborator), Gio Wielderhold (collaborator), Zoran Dimitrijevic (Phd candidate, CS), Kingshy Goh (Phd candidate, ECE), Beitao Li (Phd candidate, ECE), Yan Meng (Phd student, ECE), Raju Rangaswami (Phd candidate, CS), Gerard Sychay (MS, CS), Gang Wu (Phd candidate, ECE), Yi Wu (Phd candidate, ECE), Huaxin You (Phd candidate, Statistics).